

Basic Principles of Medicine 1

Module: Cardiovascular, Pulmonary, Renal

Lecture: CPR 17

Primary and Secondary Hemostasis

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Blood coagulation and fibrinolysis (Hemostasis)

[Coagulation Cascade Animation - Physiology of Hemostasis - YouTube](#);
[Platelet Activation and Factors for Clot Formation - YouTube](#)

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**Lippincott's Biochemistry 8th Edition – Chapter 35 (online access only,
online Q for the lecture)**

Practice questions for Hemostasis on Sakai

Concept map: Lippincott's Figure 35.25

Pre-lecture required reading and objectives

- DLA_Eicosanoids

SOM.MK.I.BPM1.1.CPR.2.BCHM. 0252	Describe the synthesis of the eicosanoid molecules, their general structure, and their precursors. Indicate inhibitors of eicosanoid synthesis.
SOM.MK.I.BPM1.1.CPR.2.BCHM. 0253	Discuss the role of eicosanoids in the regulation of inflammation (PGE2), gastric acidity, blood flow, and platelet aggregation (PGI2 and TXA2).
SOM.MK.I.BPM1.1.CPR.2.BCHM. 0254	Compare and contrast the effects of TXA2 vs TXA3 and their effects on vascular health.
SOM.MK.I.BPM1.1.CPR.2.BCHM. 0255	Discuss the role of leukotrienes, cysteinyl leukotrienes and SRS-A in asthma and in anaphylactic shock

Lecture: Primary and secondary hemostasis

SOM.MK.I.BPM1.1.CPR.2.BCHM. 0245	Outline the four phases of hemostasis.
SOM.MK.I.BPM1.1.CPR.2.BCHM. 0246	Discuss vascular spasm and the role of endothelin.
SOM.MK.I.BPM1.1.CPR.2.BCHM. 0247	Describe platelet plug formation.
SOM.MK.I.BPM1.1.CPR.2.BCHM. 0248	Explain the role of integrins GPIb, GPIa, GPIIb/IIIa and the role of von Willebrand factor in platelet plug formation
SOM.MK.I.BPM1.1.CPR.2.BCHM. 0249	Describe the role of platelet-activating factors (ADP, TXA2) and the changes in platelets on activation.
SOM.MK.I.BPM1.1.CPR.2.BCHM. 0250	Describe the coagulation cascade, including the extrinsic, intrinsic and common pathways.

Recommended Reading

- Lippincott's Biochemistry 9th Edition – Chapter 35 (Practice Questions) [Blood Clotting | Lippincott Illustrated Reviews: Biochemistry, 9e | Premium Basic Sciences | Health Library](#)
- Ref: Robbins Basic Pathology (11th Ed), Chapter 3 [Hemodynamic Disorders, Thromboembolism, and Shock - Robbins & Kumar Basic Pathology - ClinicalKey Student](#) – Section: Normal hemostasis
- Practice questions for Hemostasis on Sakai
- Concept map: Lippincott's Figure 35.25

Hemostasis and its phases

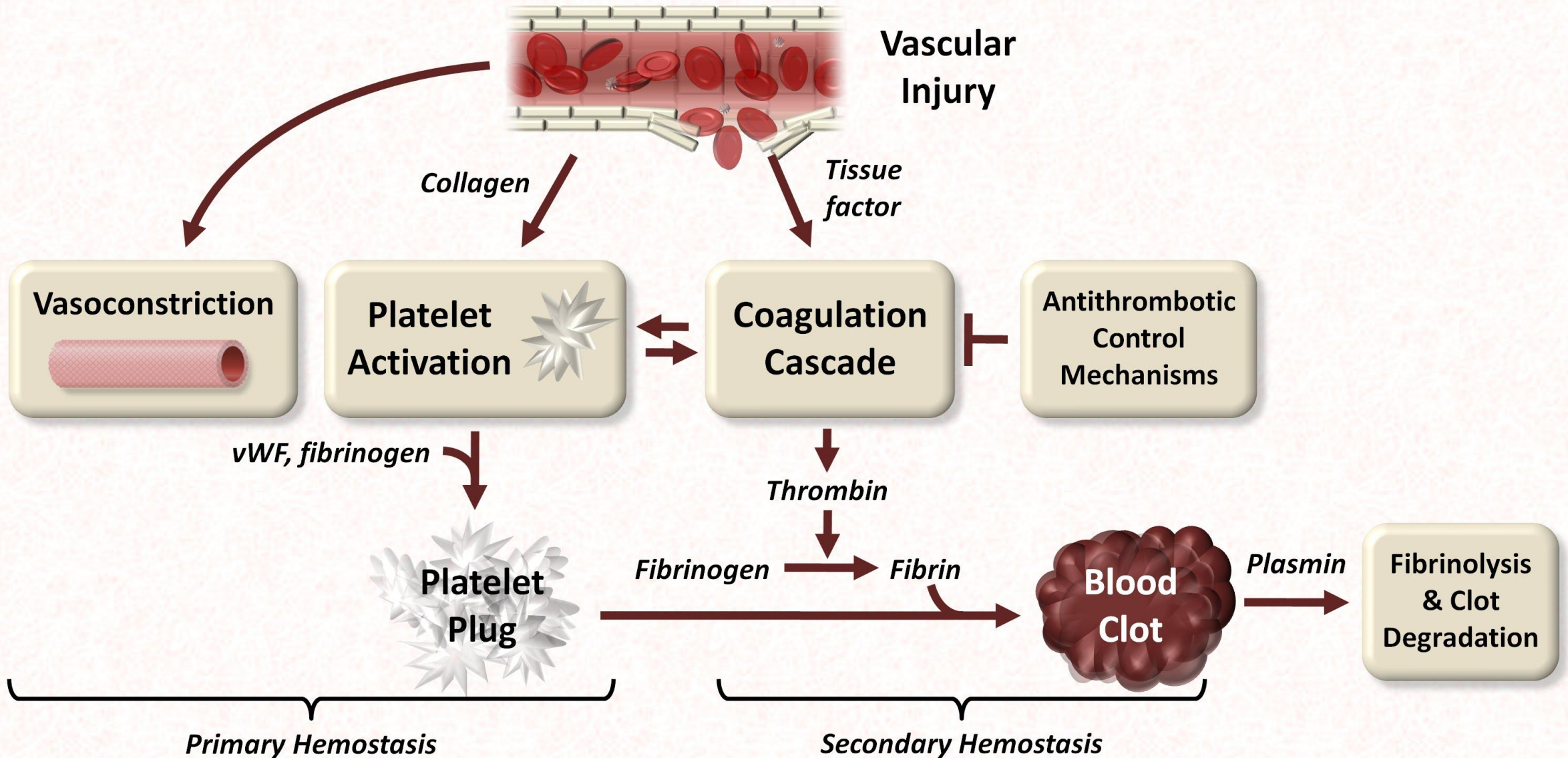
Precisely orchestrated process involving platelets, clotting factors, and endothelium

Occurs at site of vascular injury and forms a blood clot, which serves to prevent bleeding (Arrest of bleeding/ prevention of hemorrhage).

Steps in hemostasis

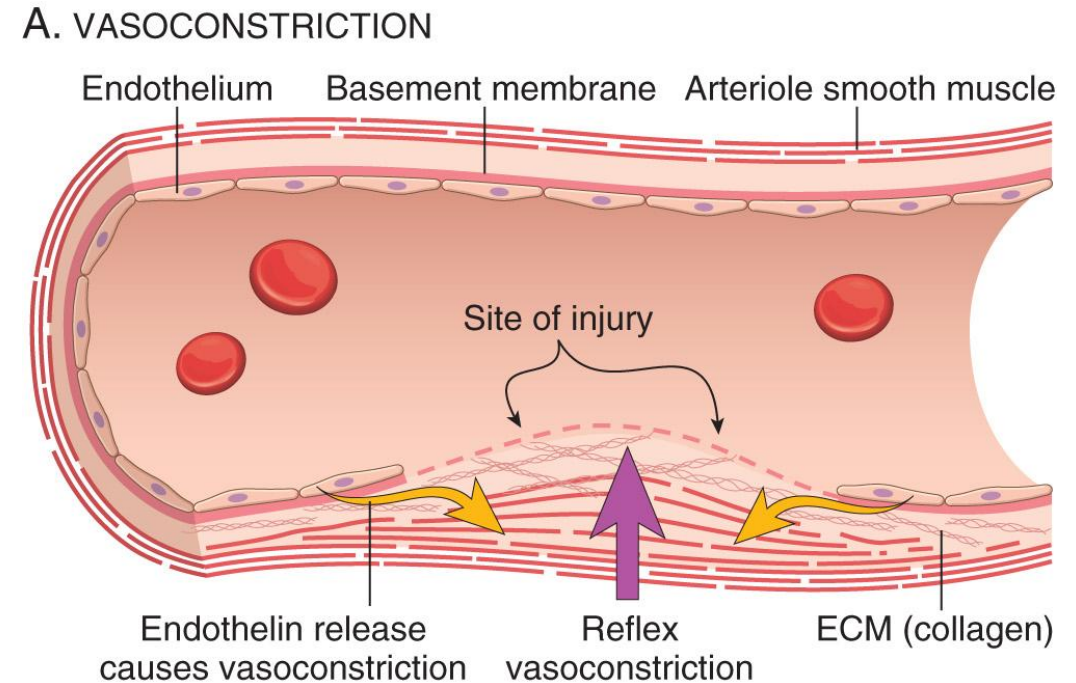
1. Vascular spasm / vasoconstriction
2. Platelet plug formation / **primary hemostasis**
3. Blood coagulation / **secondary hemostasis**
4. Clot stabilization and resorption

Major Components of Hemostasis



1. Vascular spasm

- Trauma to vessel wall results in smooth muscle contraction
- Contraction caused by
 - Local myogenic spasm
 - Factors released from injured vessel wall (endothelin, serotonin when endothelial lining is damaged)
 - Nervous reflexes



Vasoconstriction is transient and cannot cause long term cessation of bleeding

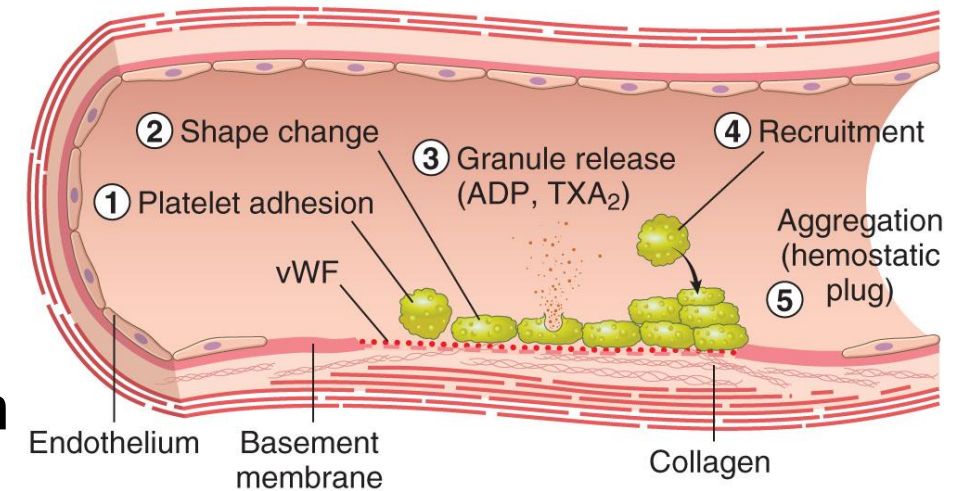
2. Platelet plug formation: Primary hemostasis

- Greatly limits loss of blood from site of injury by forming plugs
- Small cuts in blood vessels often sealed by platelet plugs
- **Lab tests: Bleeding time and platelet aggregation tests**

- **Stages of platelet plug formation**

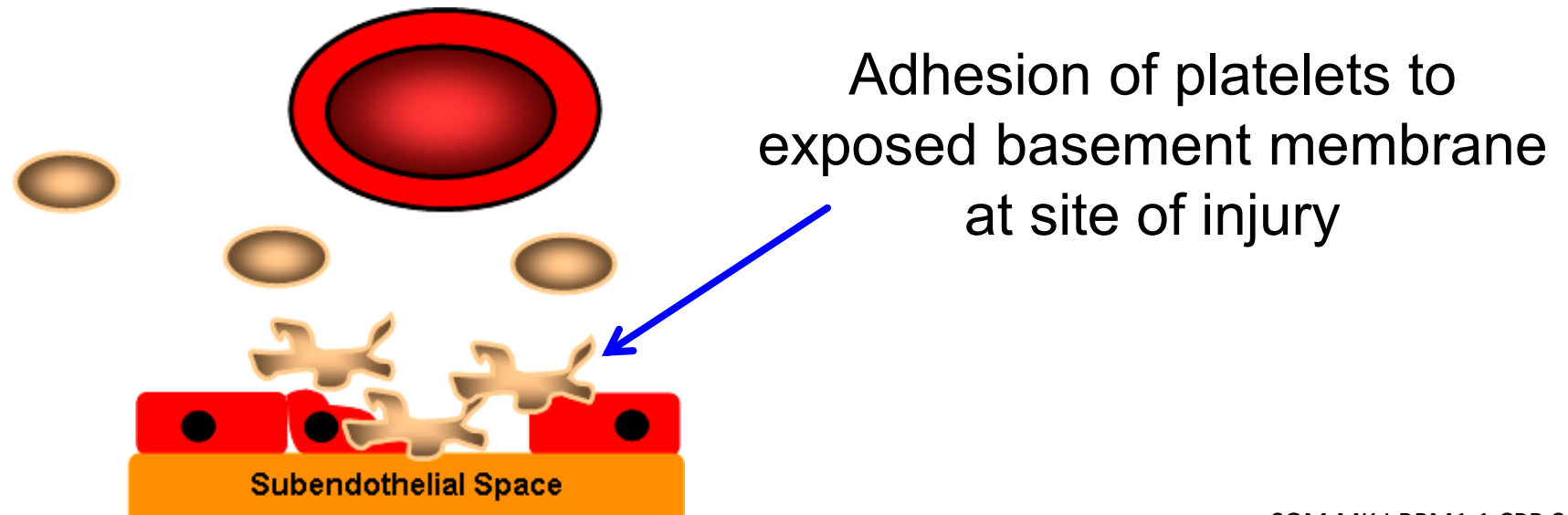
- **Platelet adhesion** to the damaged surface
- **Platelet activation**
- **Platelet recruitment and Platelet aggregation**

B. PLATELET ACTIVATION AND AGGREGATION

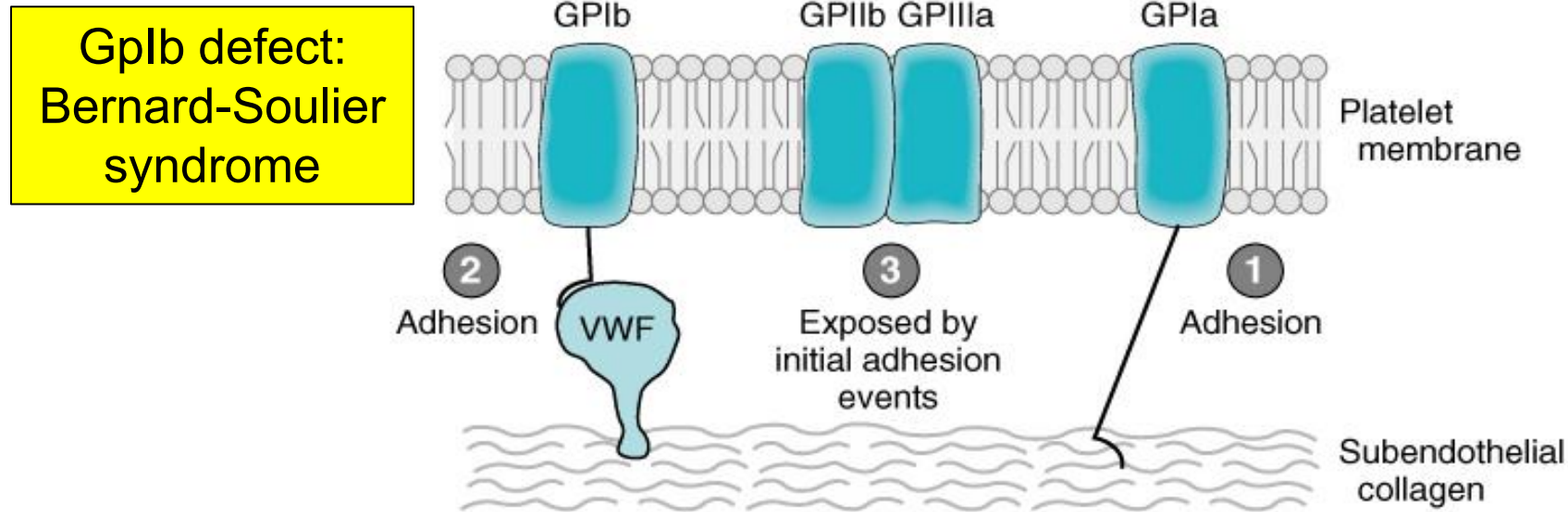


Platelet adhesion

- Endothelial injury facilitates platelet adherence and activation (Prevented under normal circumstances by negative charges on both platelets and endothelial cells)
- Mediated by platelet receptors: **Gp-Ia and Gp-Ib**
- Receptors bind to subendothelial matrix e.g. collagen, vWF

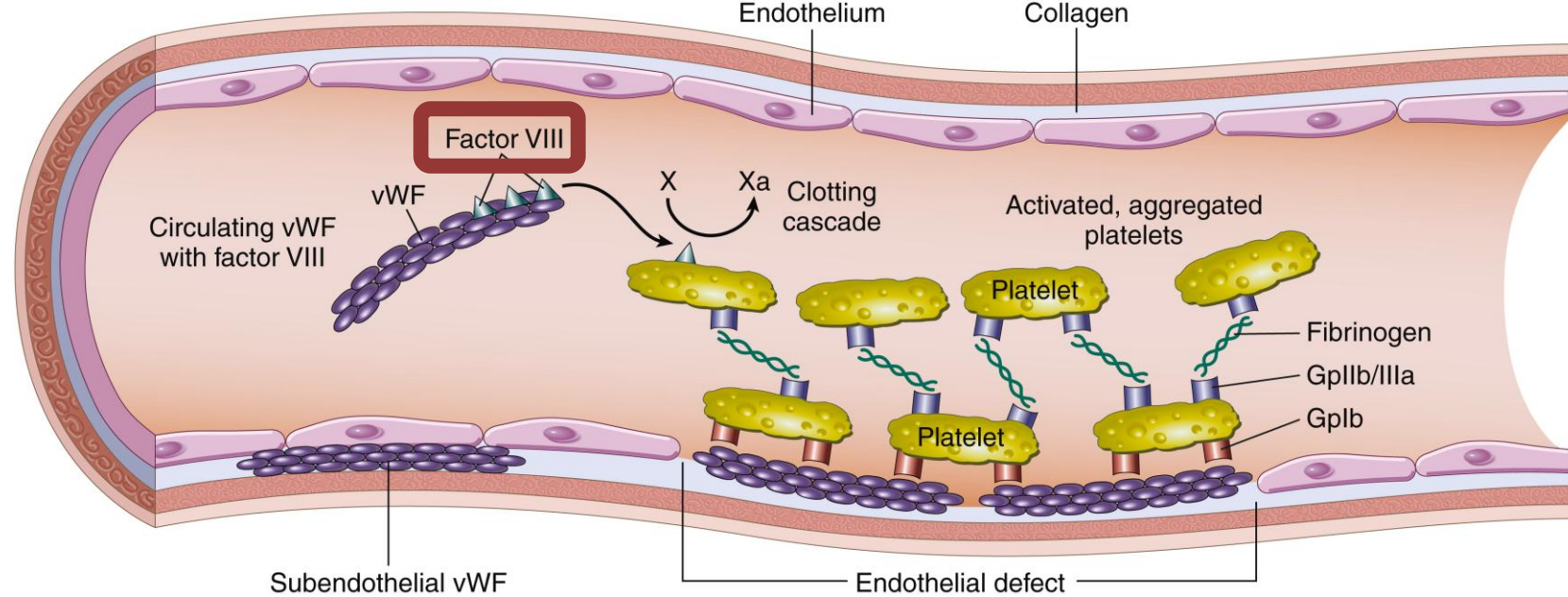


Platelet adhesion and role of von Willebrand factor (vWf)



1. Platelet **GPIa** binds **collagen** → Structural changes like development of pseudopods to promote platelet-platelet interactions
2. **Von Willebrand factor** binds to platelet receptor, **Glycoprotein Ib** (GPIb) resulting in changes in platelet membrane
3. Binding also exposes **GPIIb/IIIa** for binding of **fibrinogen**

von Willebrand factor (vWF): Platelet adhesion and coagulation cascade



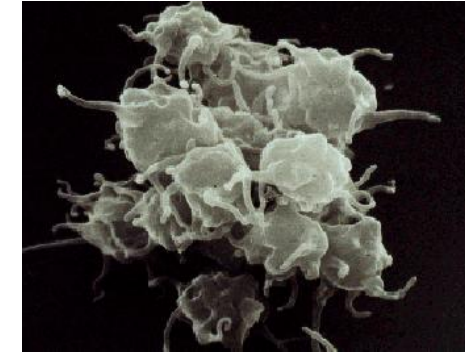
- vWF: Weibel-Palade bodies in endothelial cells (Histology)
- vWF: Bridge between platelet glycoproteins (GPIb) and collagen fibers;
Facilitates platelet adhesion to vessel wall and aggregation
- vWF complexes with **factor VIII**; **Stabilizes** and prevents its degradation (Secondary hemostasis)
- vWf deficiency associated with
 - Defect in platelet plug formation (primary hemostasis)
 - Defect in coagulation cascade (factor VIII unstable) → defect in secondary hemostasis

Immediately following initial platelet adhesion



Normal discoid shape

Platelet Activation: Shape change



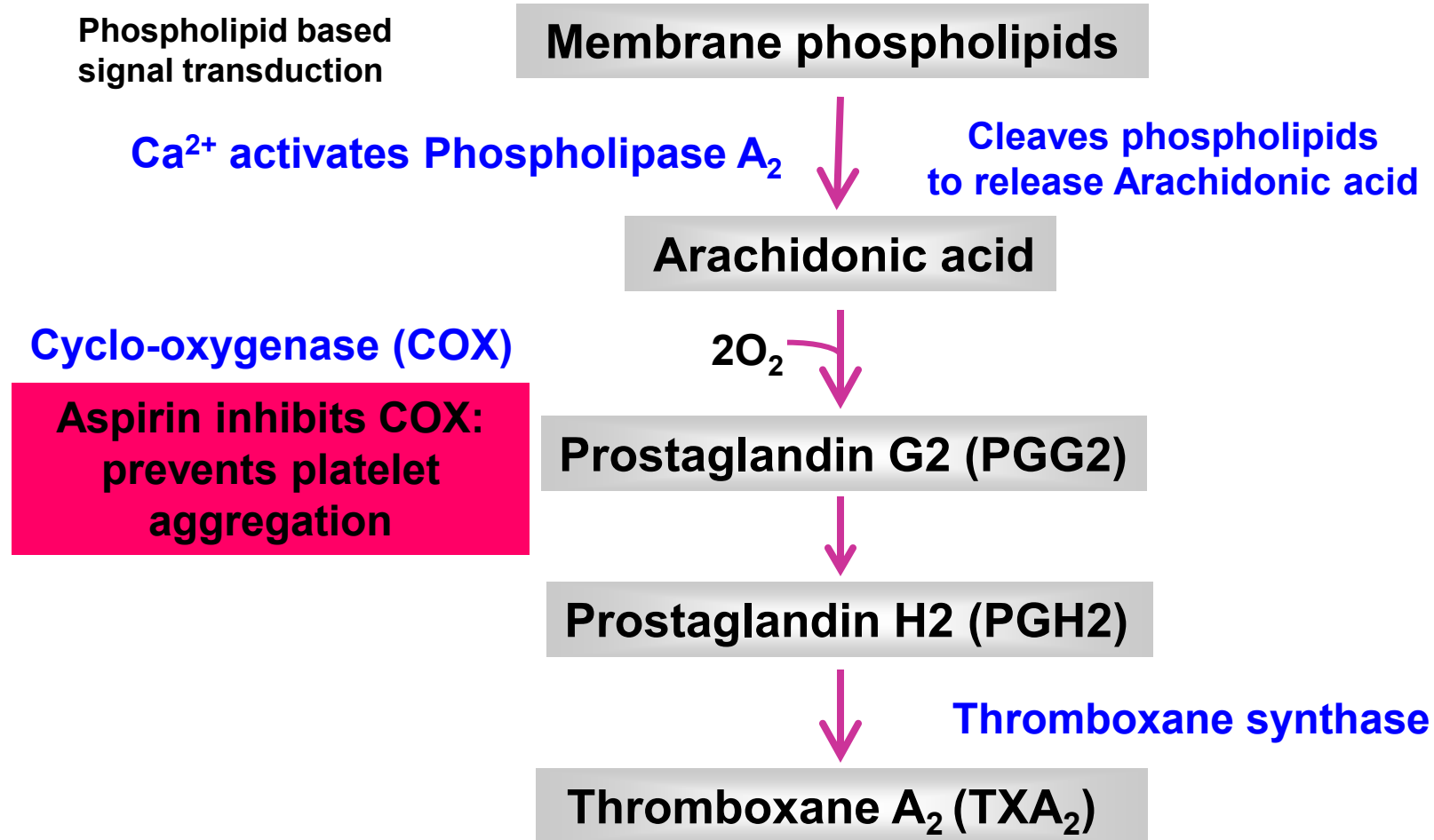
Activated platelets

Degranulation (**dense granules ADP**),
Activates platelets, promotes aggregation

**Increases
Thromboxane A₂**

- **ADP** binds to neighboring platelets
 - **Increases intracellular Ca²⁺** and
 - **Decreases intracellular cAMP levels**
(Facilitates platelet aggregation)
- ***Platelet Aggregation cannot take place without ADP stimulation***

↑ Intracellular Ca^{2+} stimulates platelets to synthesize TXA_2

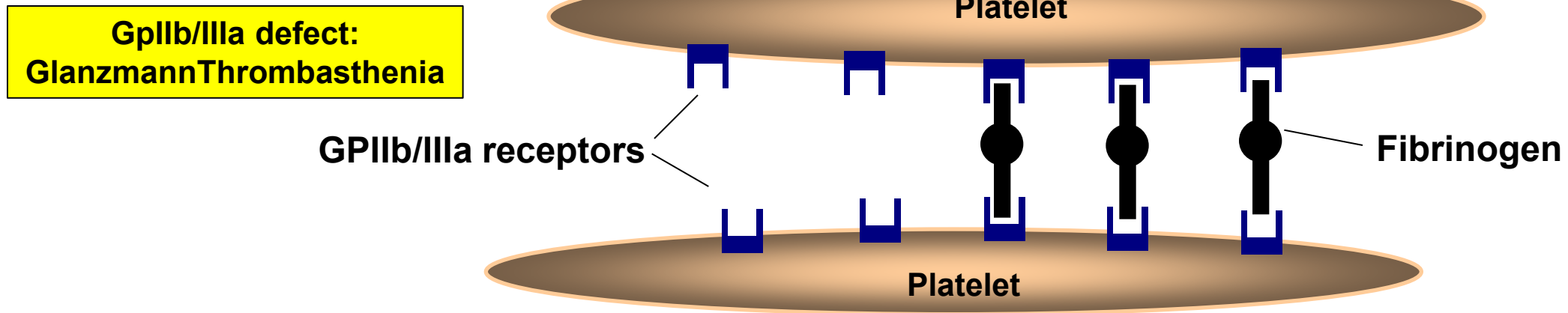


Review DLA: Eicosanoids

TXA_2 induces platelet aggregation by activating other platelets;
Enhances vasoconstriction.

Platelet aggregation

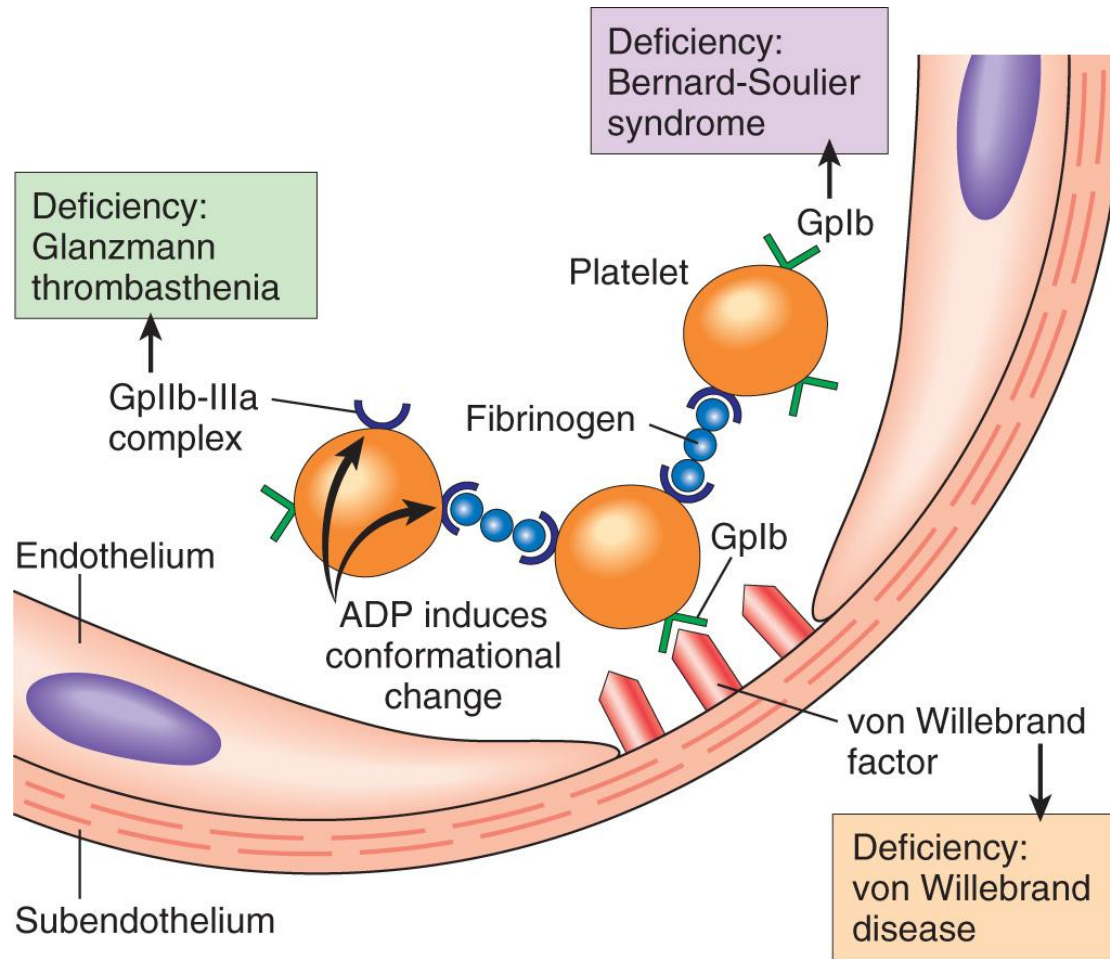
Platelet aggregation mediated by **fibrinogen**.
Fibrinogen binds to GPIIb/IIIa on adjacent platelets



Formation of platelet plug/ primary hemostasis

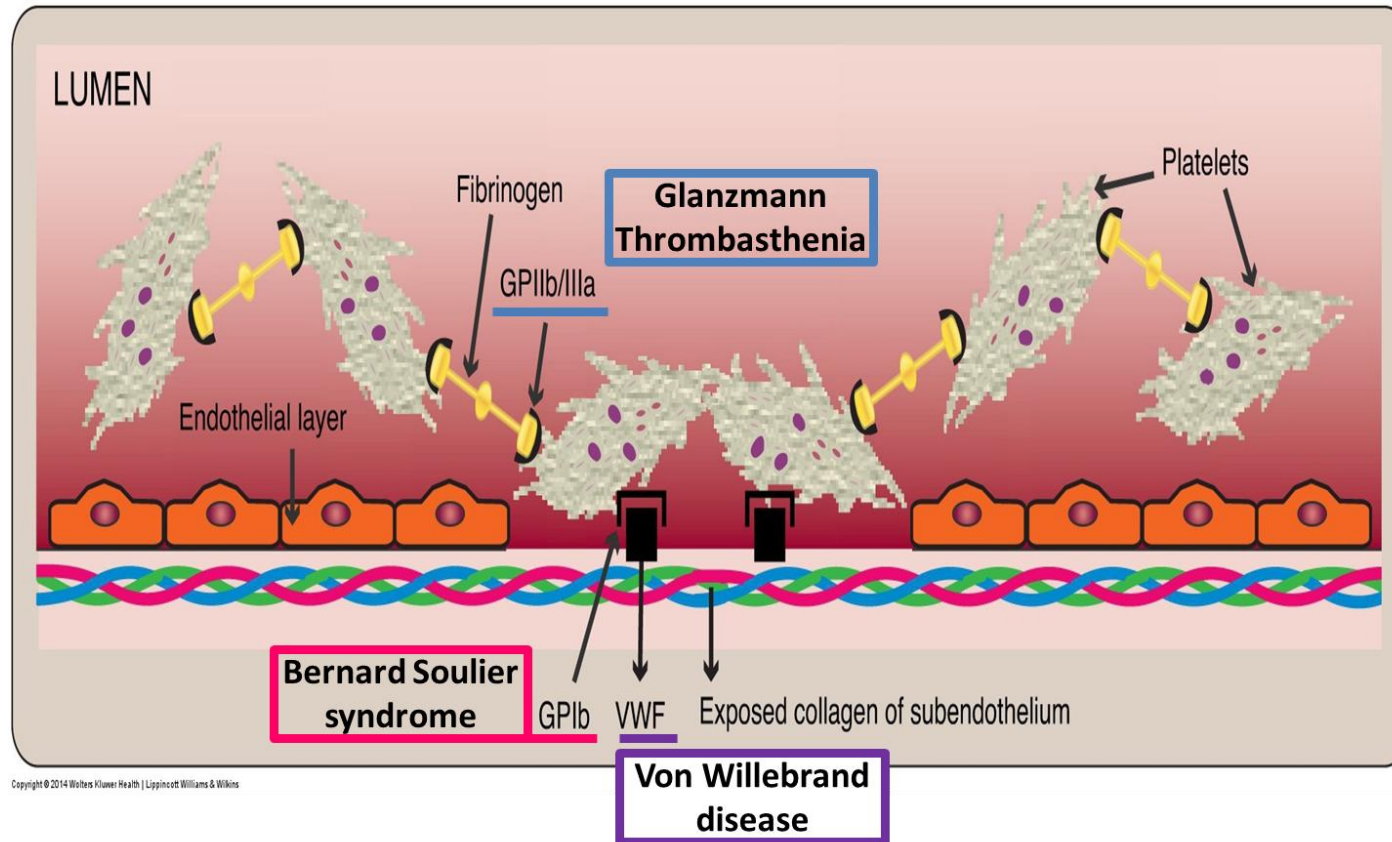
Formation of platelet plug (Primary hemostasis)

<https://www.youtube.com/watch?v=R8JMfbYW2p4>



1. Subendothelial collagen exposure
2. **Platelet adhesion** via GPIa and GPIb (via **vWF**) to subendothelial collagen
3. **Platelet activation**, shape change and **degranulation** (**ADP** release)
4. **ADP** binds to neighboring platelets to increase intracellular calcium and decrease intracellular cAMP; Increased TXA₂
5. Platelets recruited to injury site
6. Platelet aggregation via fibrinogen linking adjacent platelets via GPIIb/IIIa

Primary hemostasis: Formation of platelet plug



Primary hemostasis defect:
Prolonged bleeding time
Defective platelet aggregation

LAB TESTS: Platelet plug formation defects result in **prolonged bleeding time** and **defective platelet aggregation** with stimulants

Formation of platelet plug requires von Willebrand factor, adequate number of platelets, intact GpIb and GPIIb/IIIa platelet receptors.

Tests for platelet function/aggregation

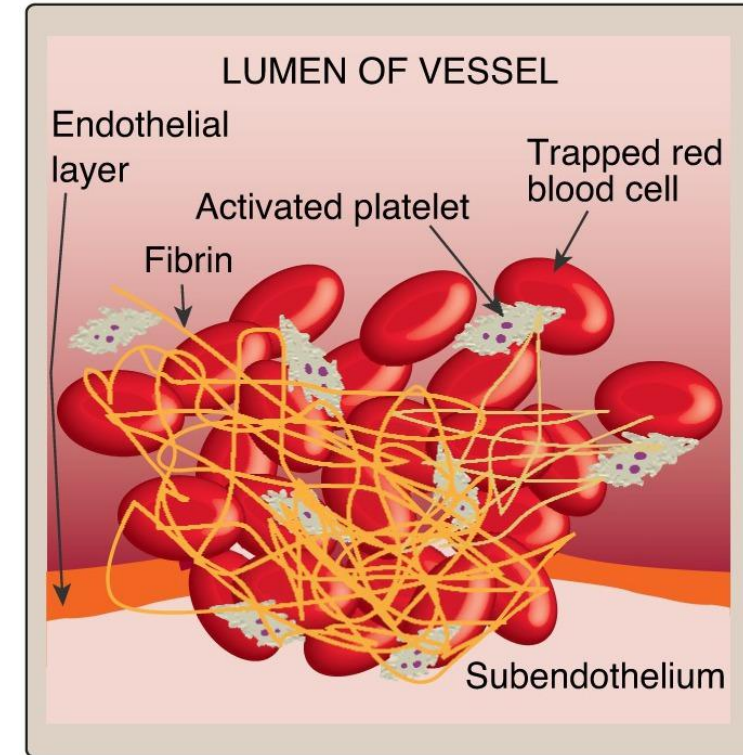
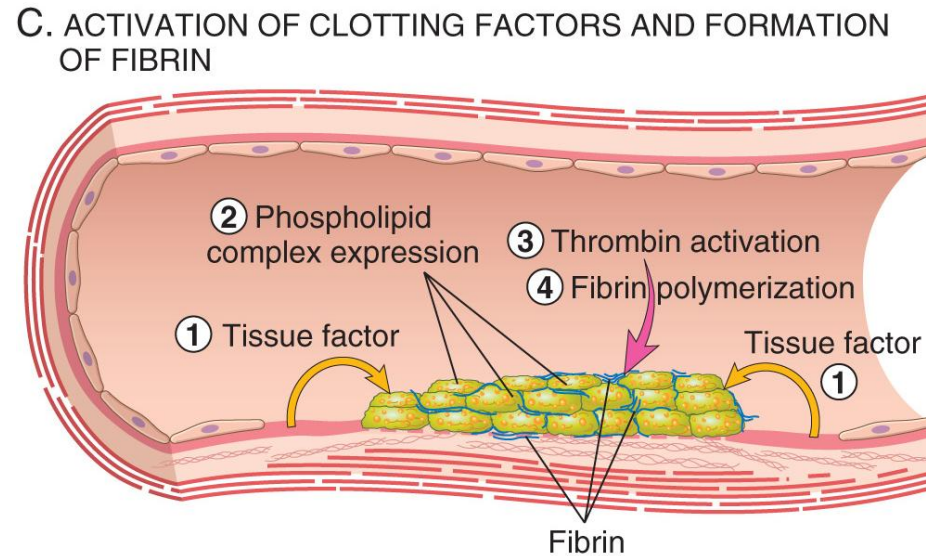
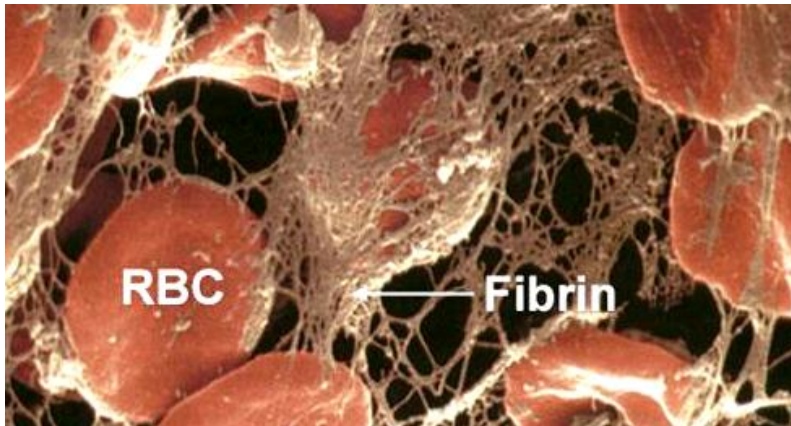
Bleeding time (BT)

- Time taken from **initial injury to formation of platelet plug**
- Indicator of **platelet plug formation (primary hemostasis)**
- **Not a reliable test**
- Prolonged bleeding time is an indicator of
 - Low platelet count OR
 - vWf deficiency OR
 - Platelet receptor defects

- **Platelet count**
- **Platelet aggregation/ function tests** measure platelet aggregation in presence of activators (ADP, thrombin, and ristocetin)
- **Ristocetin cofactor assay:** von Willebrand disease. Reduced platelet aggregation in von Willebrand disease
- **Flow cytometry:** Platelet glycoprotein defects

3. Blood coagulation (secondary hemostasis)

“Conversion of blood from liquid state to a **solid, gel like state**”



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Aim: Convert **soluble fibrinogen to insoluble fibrin** (Stabilizes platelet plug)
Conversion of fibrinogen to fibrin requires thrombin

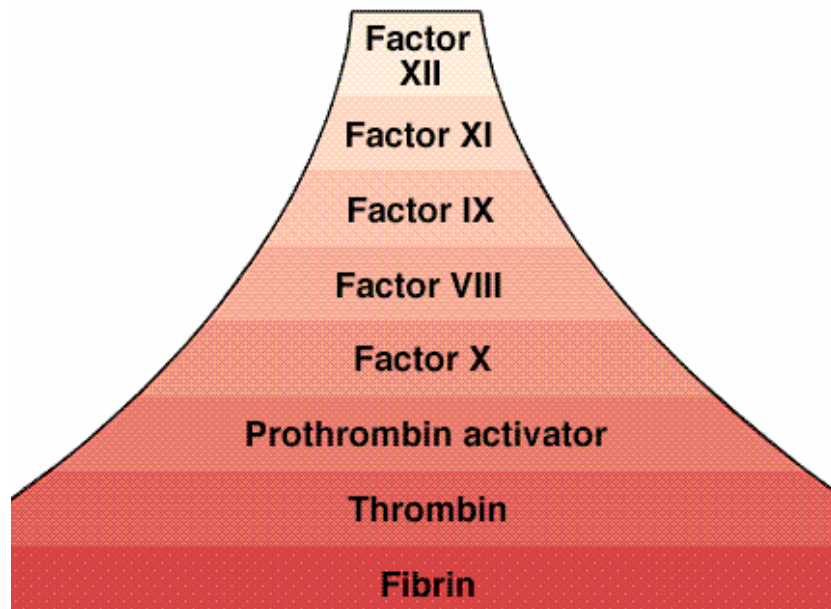
Production of thrombin: Coagulation cascade

Thrombin formation occurs via 2 pathways:

- 1) Intrinsic pathway
- 2) Extrinsic pathway

<https://www.youtube.com/watch?v=MPGe-guZMqM>

Enzyme Amplification

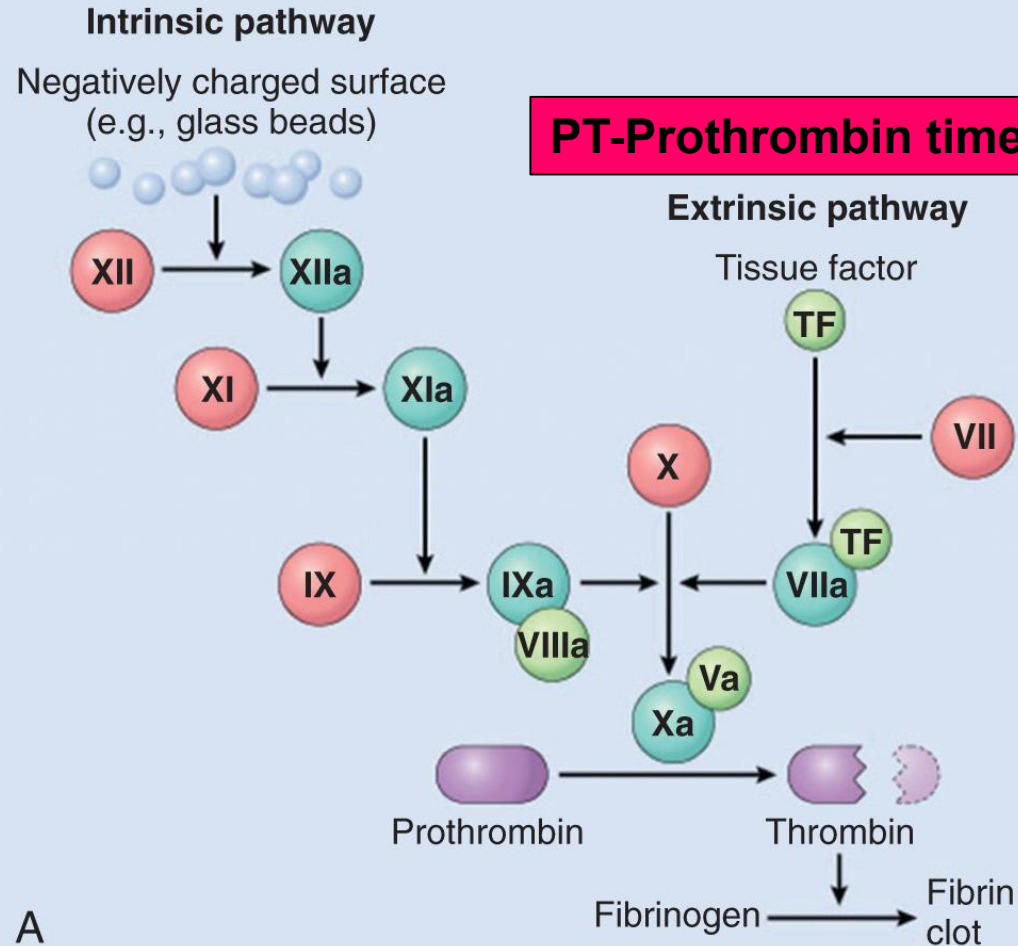


- Cascade of enzymatic reactions
- Transform proenzymes to their active forms (**irreversible proteolytic activation**)
- Most clotting factors present in circulation in an inactive form
- Product formed at each step catalyzes the next step

Coagulation cascade: Memorize pathway

APTT

CLOTTING IN THE LABORATORY

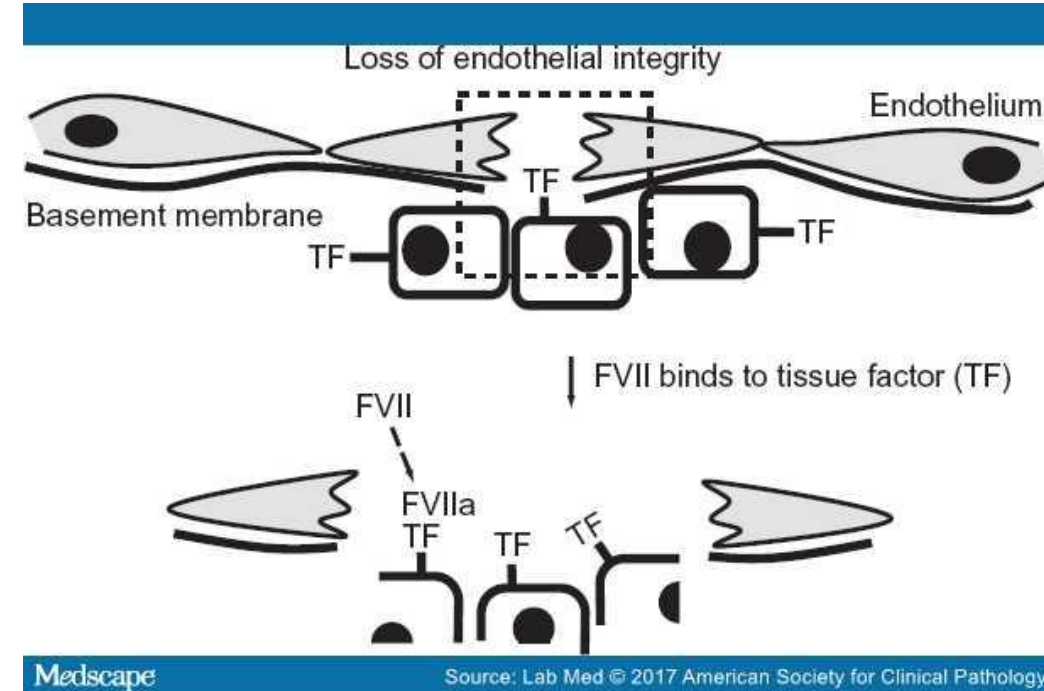
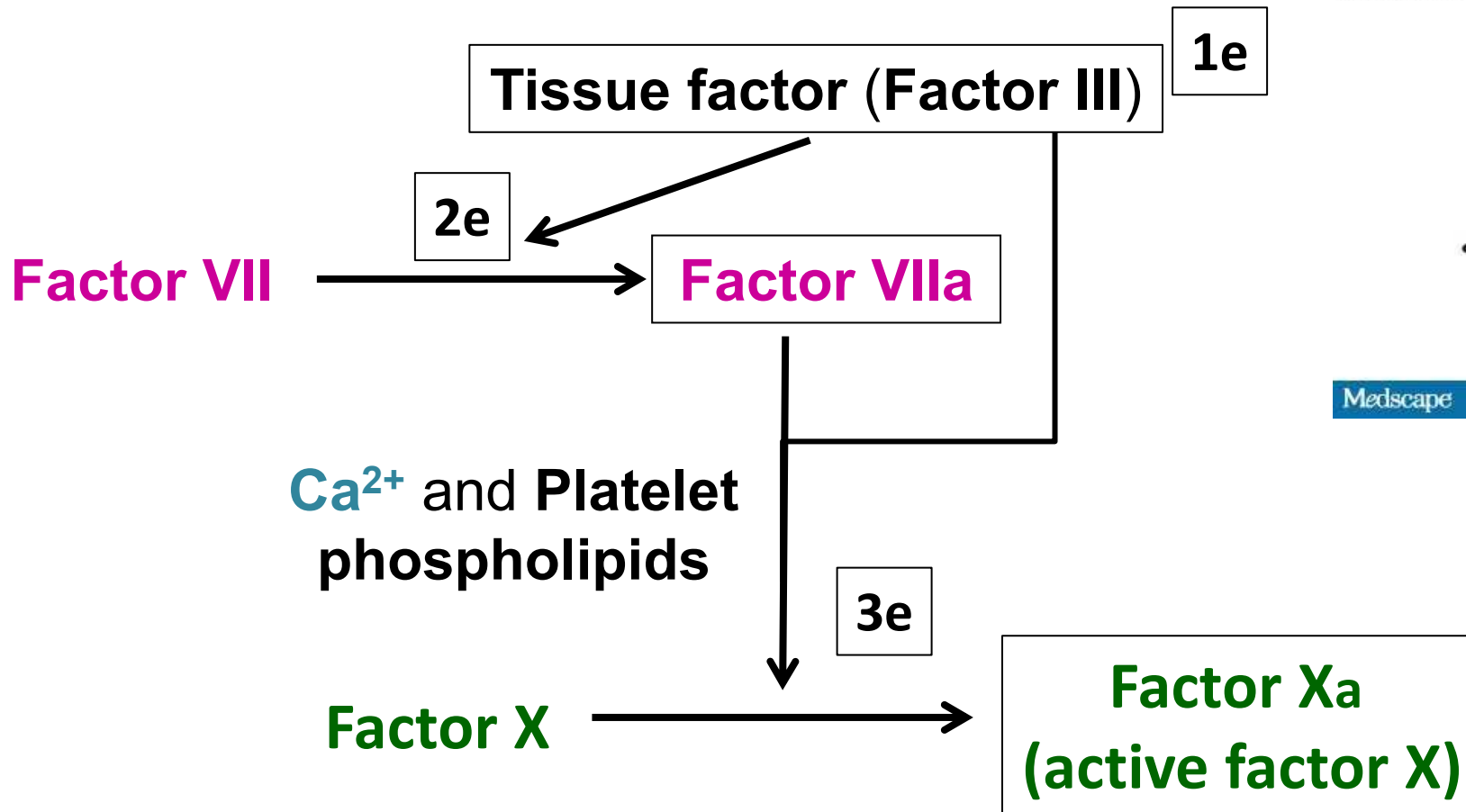


A

- Factor X activation
 - Extrinsic pathway
 - Intrinsic pathway
- Factor X forms thrombin
- Thrombin forms FIBRIN

Extrinsic pathway for activation of factor X

Vascular damage exposes **Tissue factor**



Extrinsic pathway for activation of factor X

**PT - Prothrombin time/
INR - International normalized ratio**

Tissue injury → release of
Tissue factor (Factor III) 1e



Activates Factor VII 2e



Factor VII_a and tissue factor, Ca²⁺ and
platelet phospholipids, activate factor X



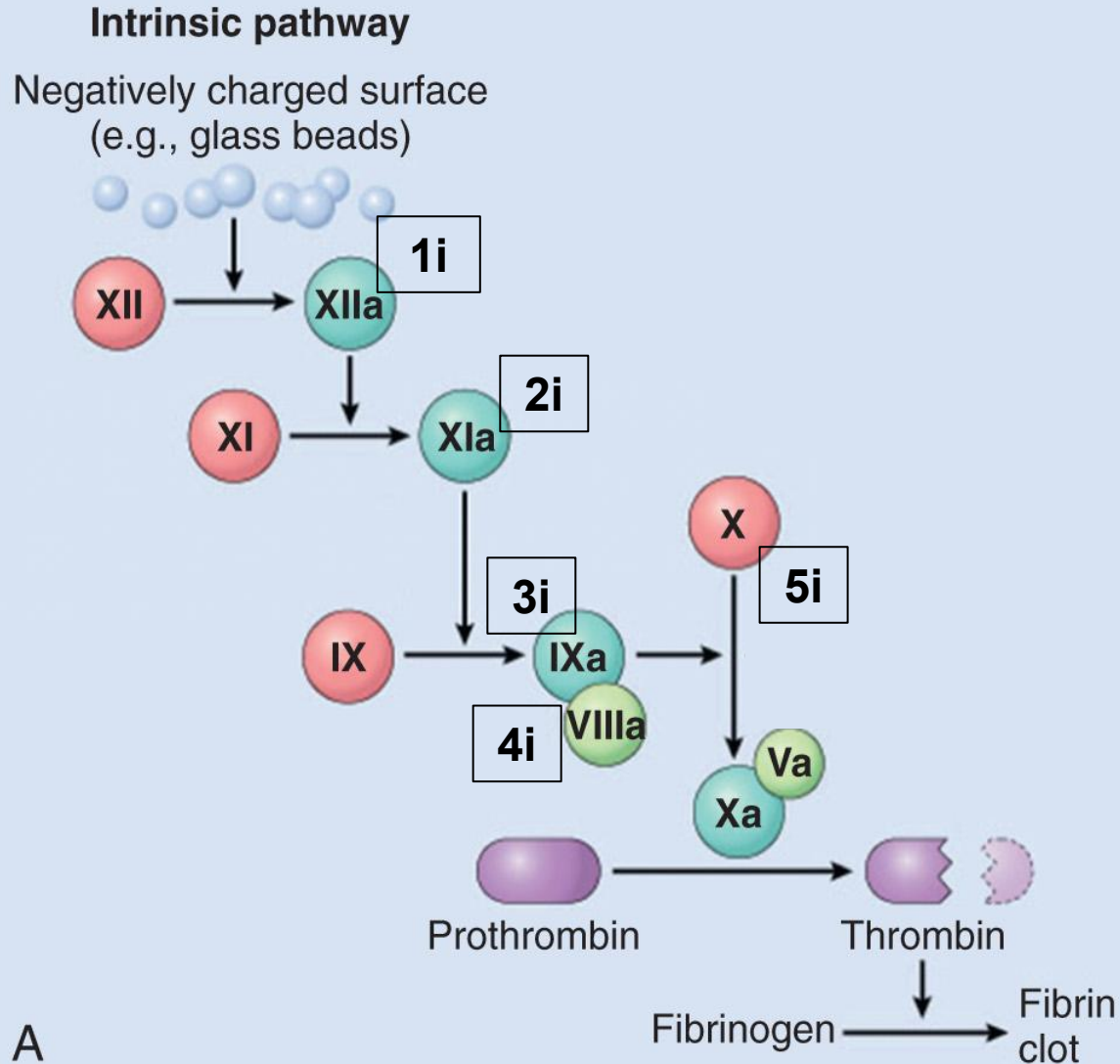
3e

X_a (active factor X)

Intrinsic pathway for activation of factor X

APTT - Activated Partial Thromboplastin time

CLOTTING IN THE LABORATORY



Activation of factor XII 1i

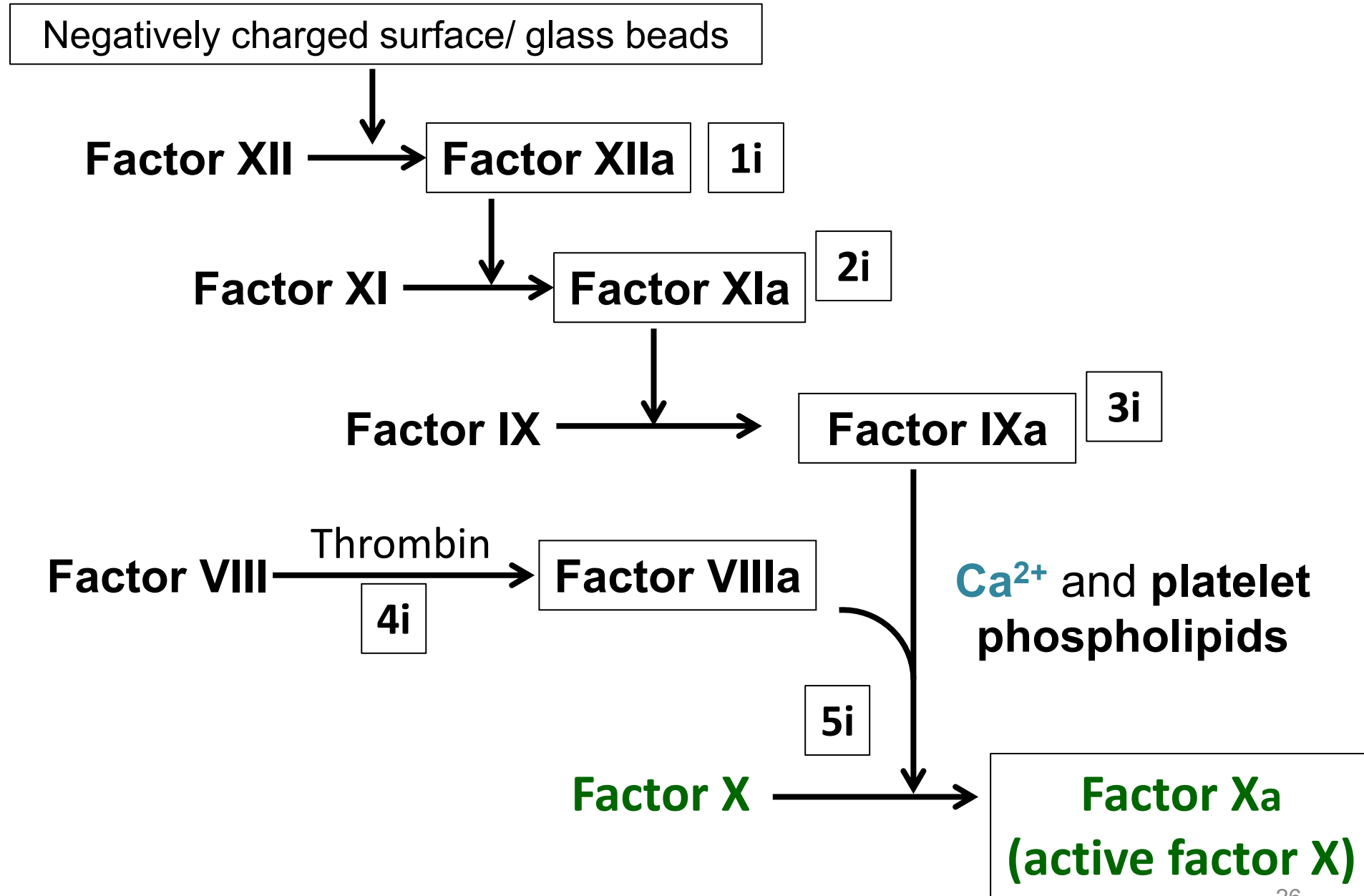
Activation of factor XI 2i

Activation of Factor IX 3i

Thrombin activates factor VIII → VIIIa 4i

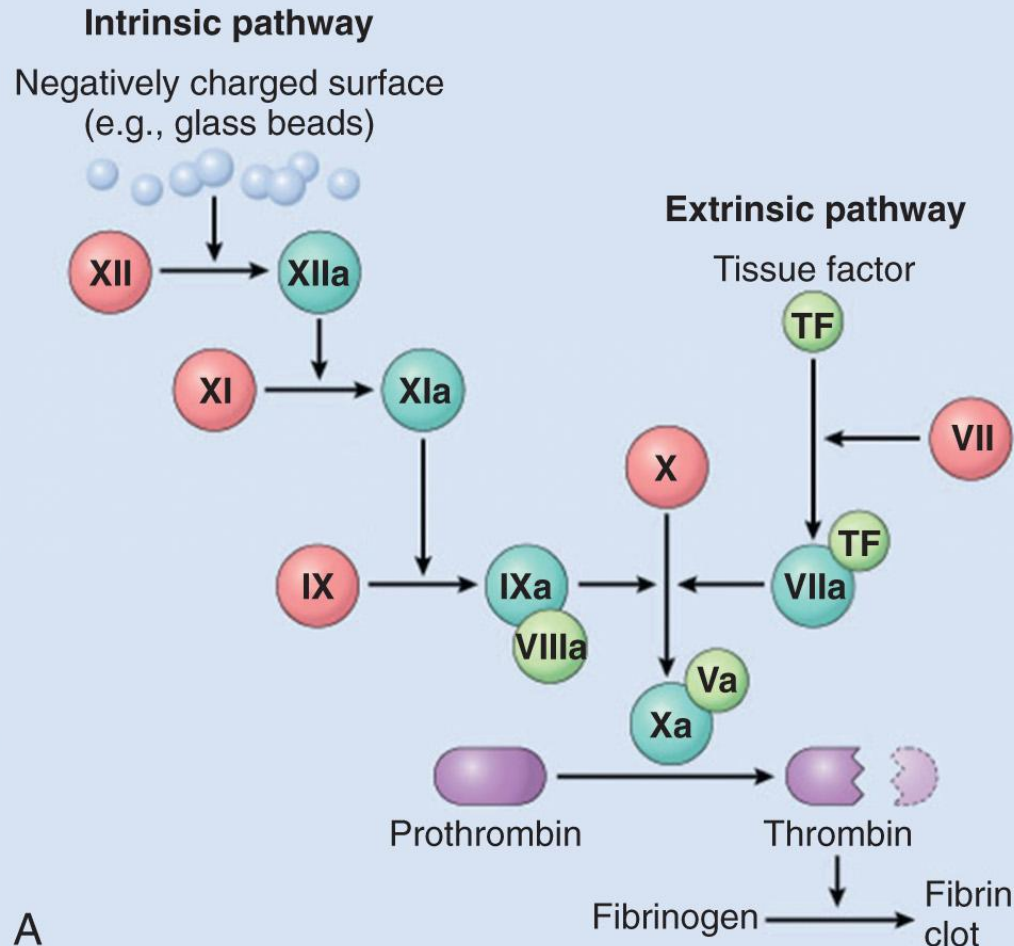
IXa, VIIIa, platelet phospholipids and Ca^{2+} , activate factor X to Xa 5i

Intrinsic pathway for activation of factor X



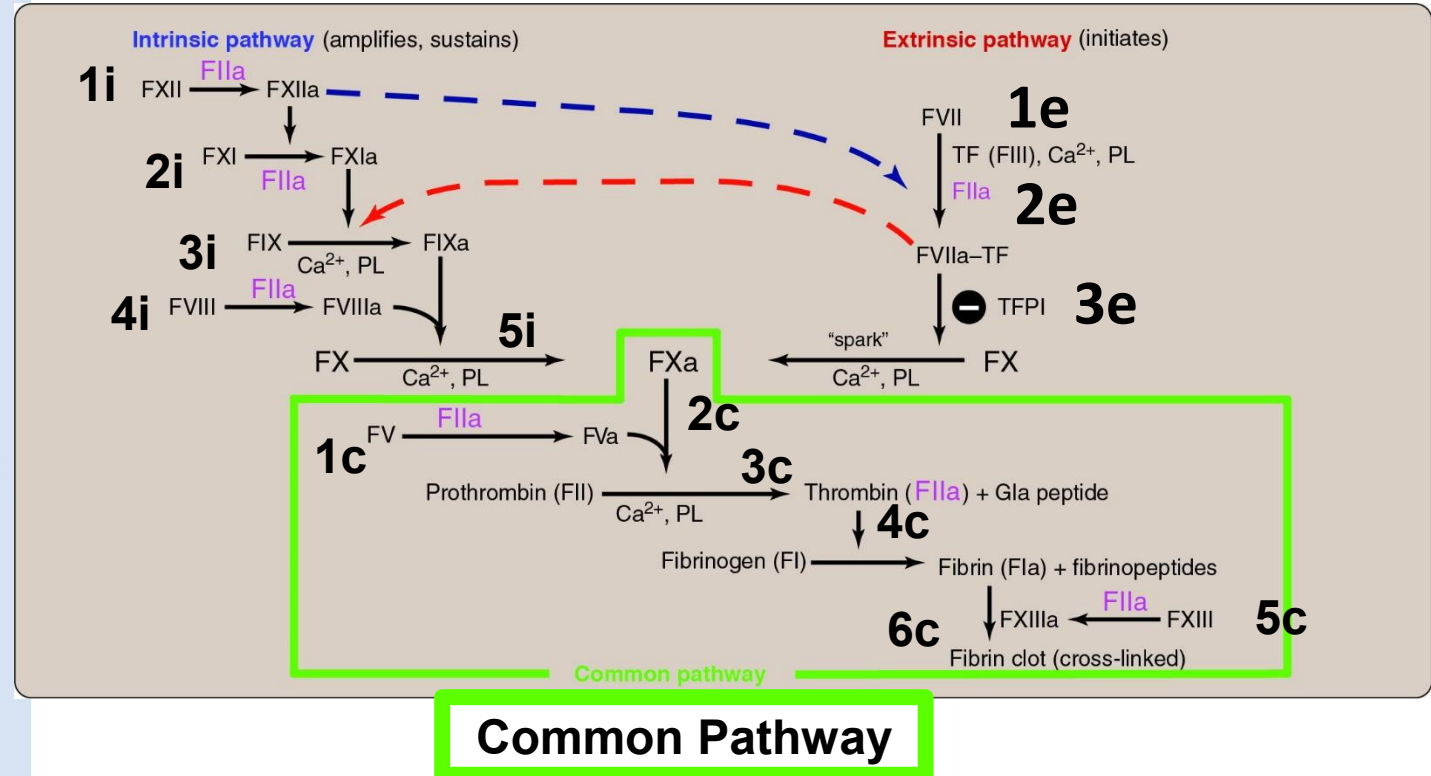
Coagulation cascade

CLOTTING IN THE LABORATORY

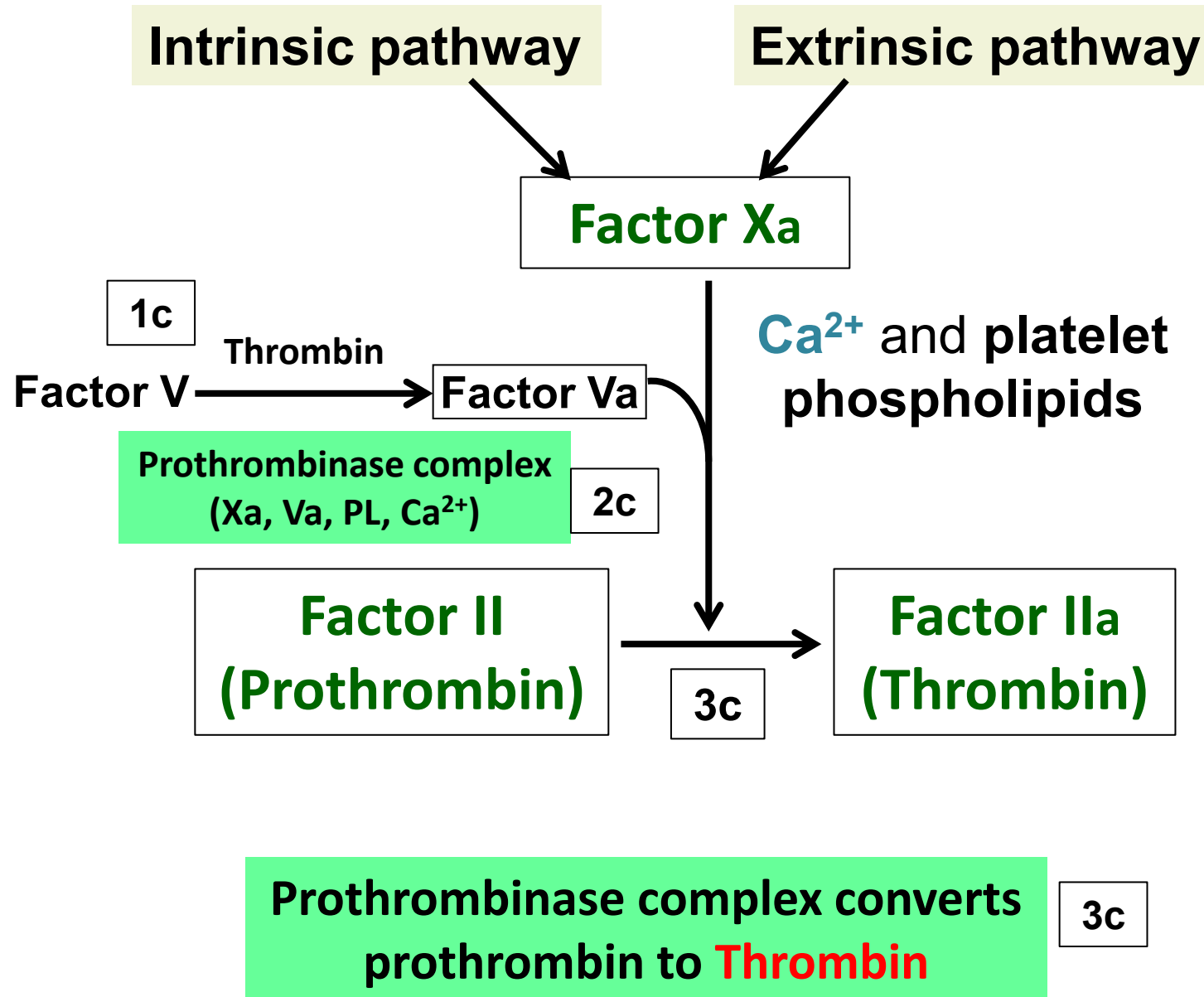


APTT

PT-Prothrombin time/ INR



Common pathway of clotting: **Formation of Thrombin**

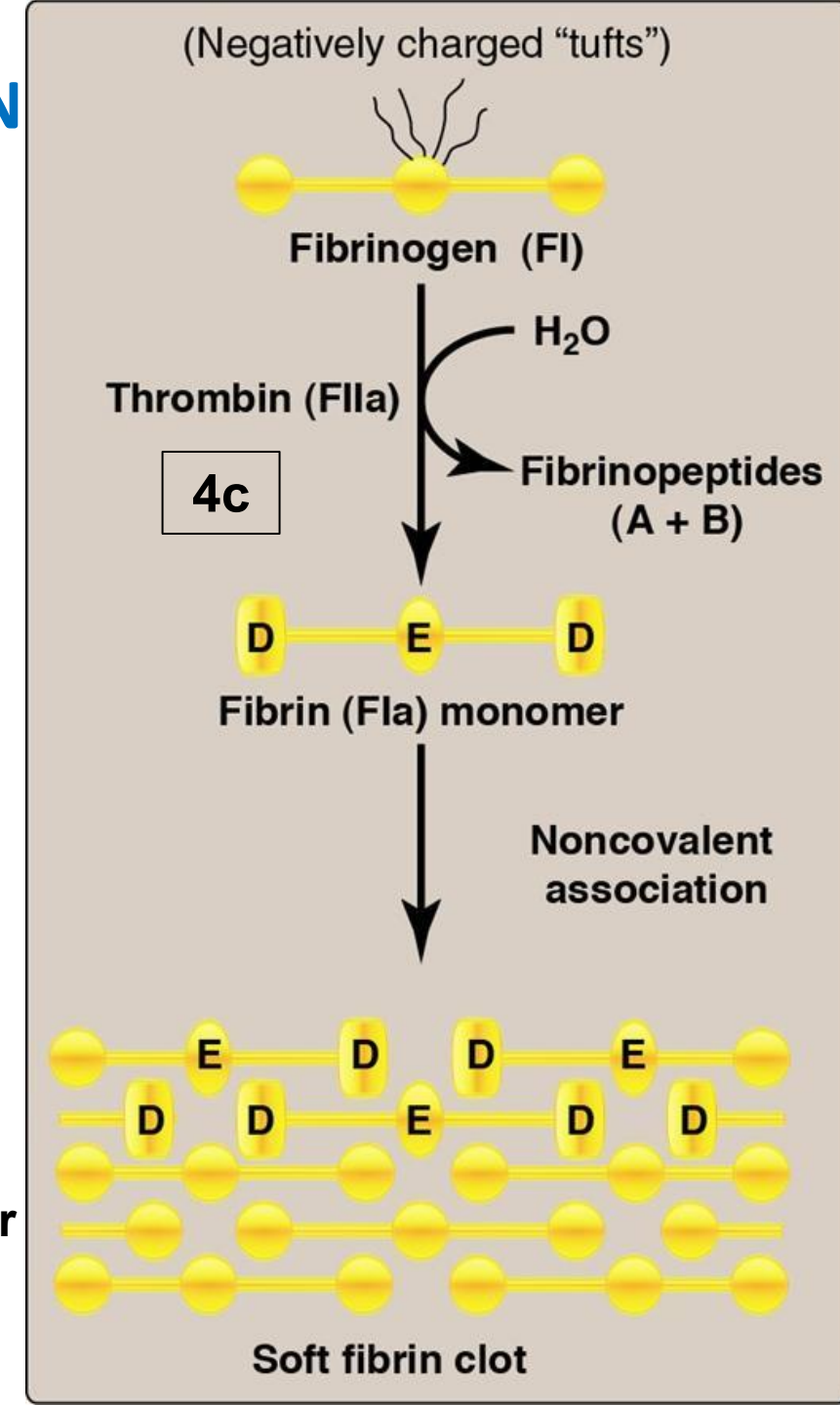


Conversion of soluble fibrinogen into insoluble FIBRIN

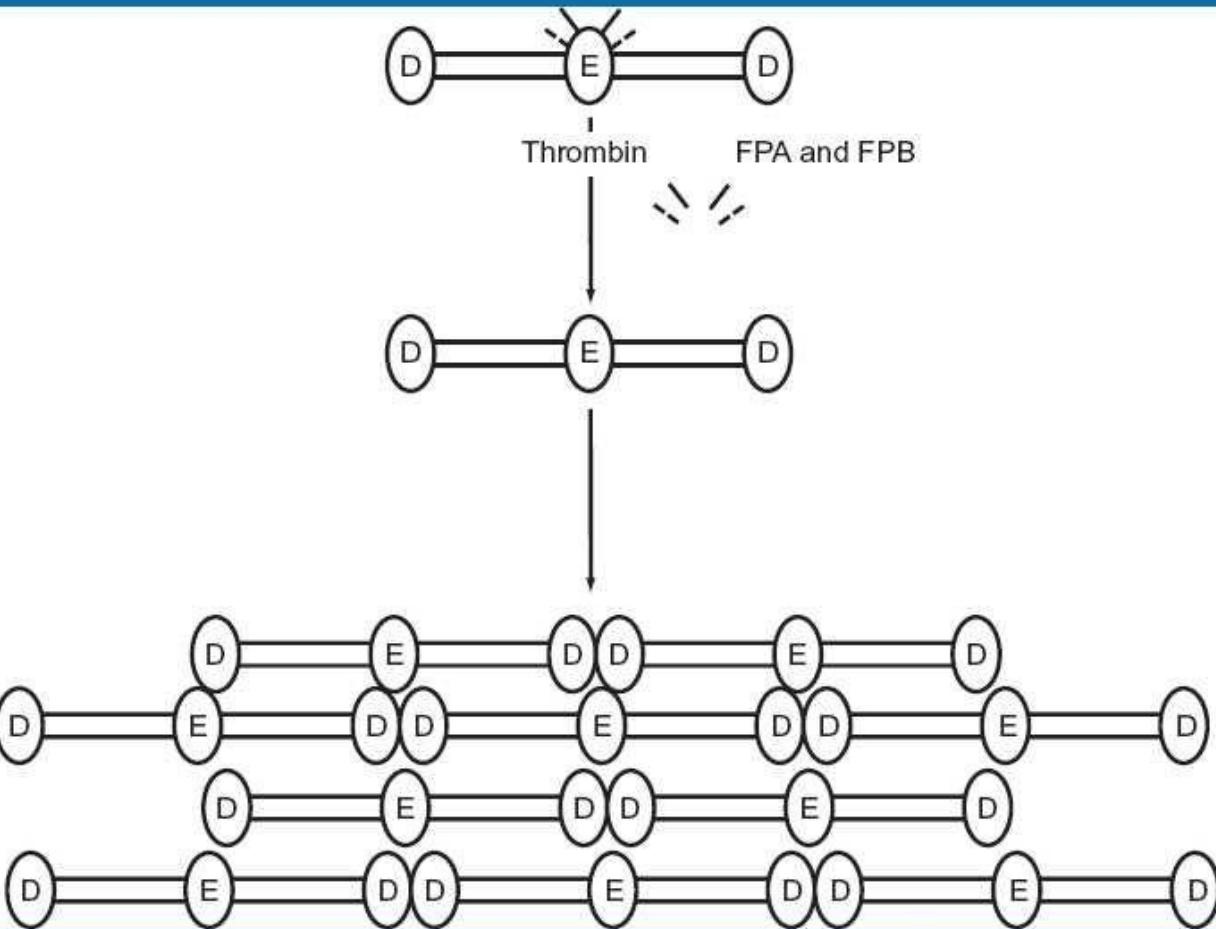
Fibrinogen (plasma protein)
synthesized by liver

4c Thrombin cleaves fibrinogen to
form the fibrin monomer

Fibrin monomers aggregate and linked
via **hydrogen bonds**, forming fibrin
polymer (**soft clot**)



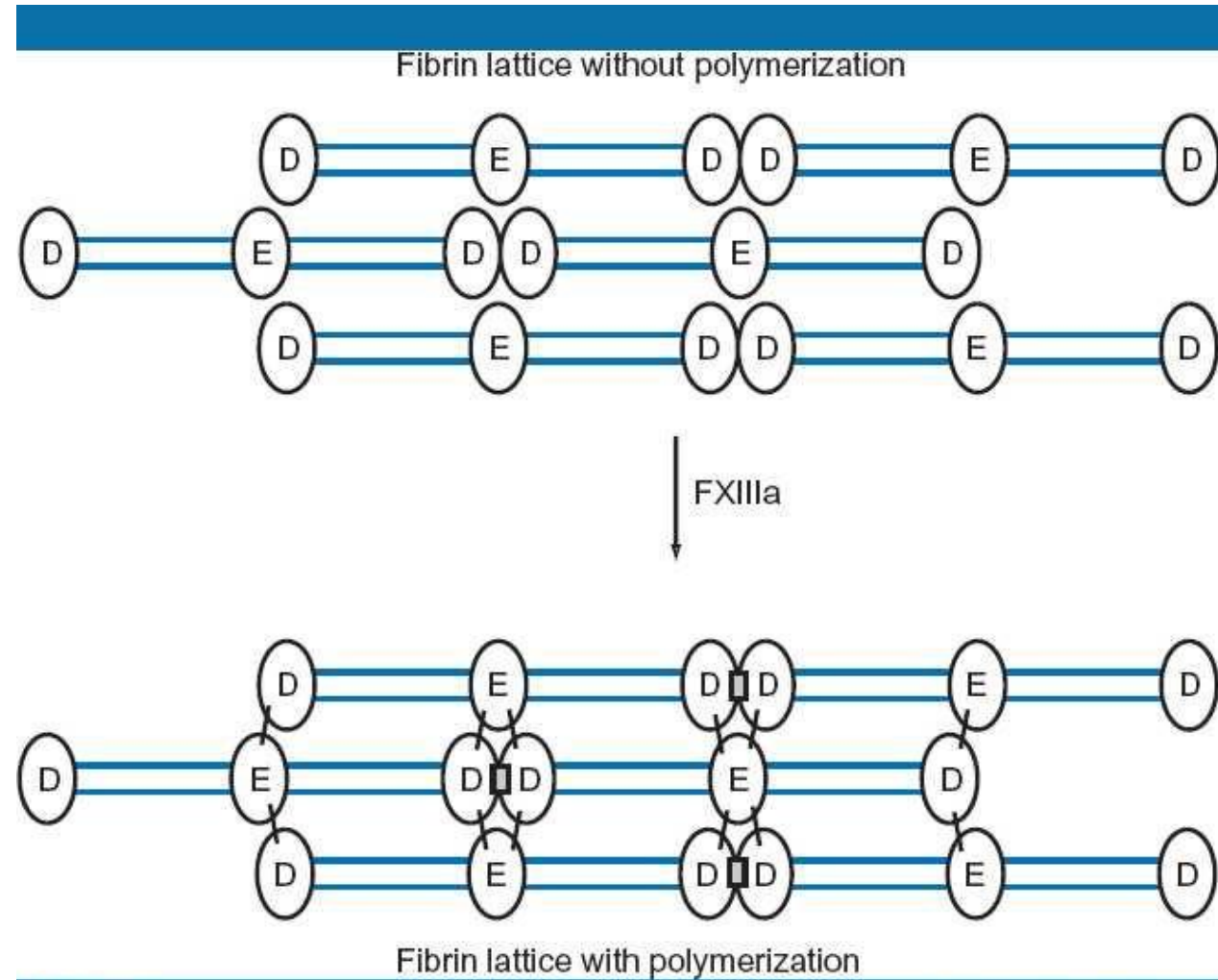
Conversion of soluble fibrinogen into insoluble fibrin by thrombin



Medscape

Source: Lab Med © 2017 American Society for Clinical Pathology

Non-covalent links

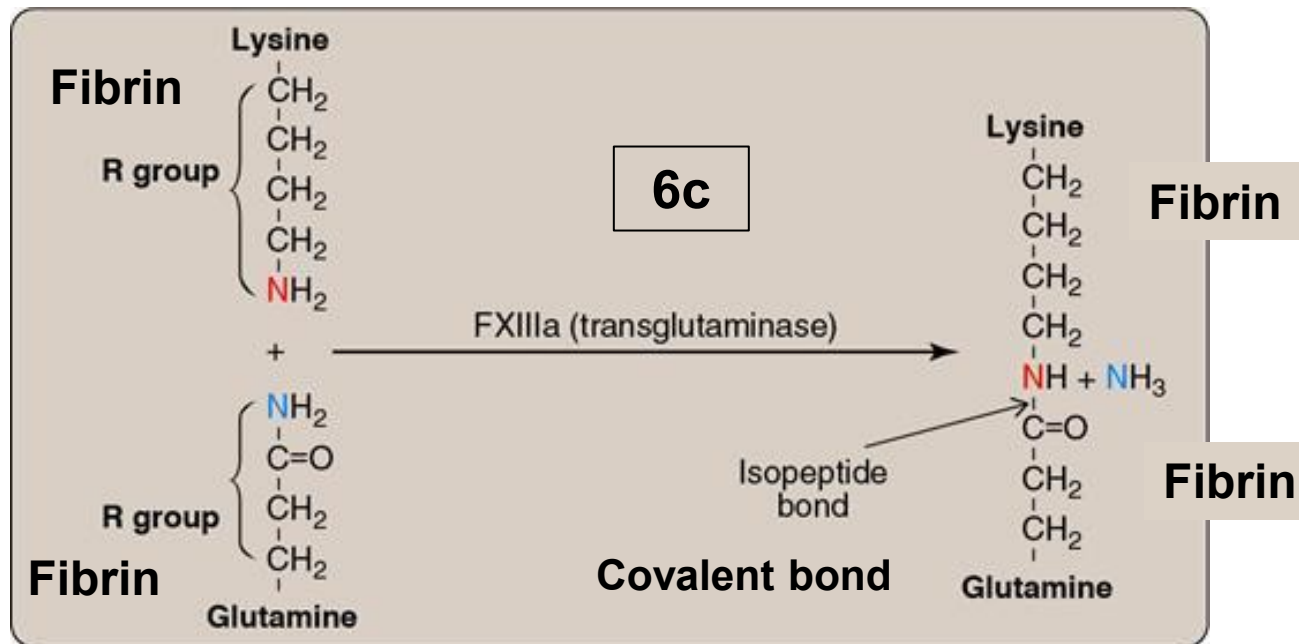
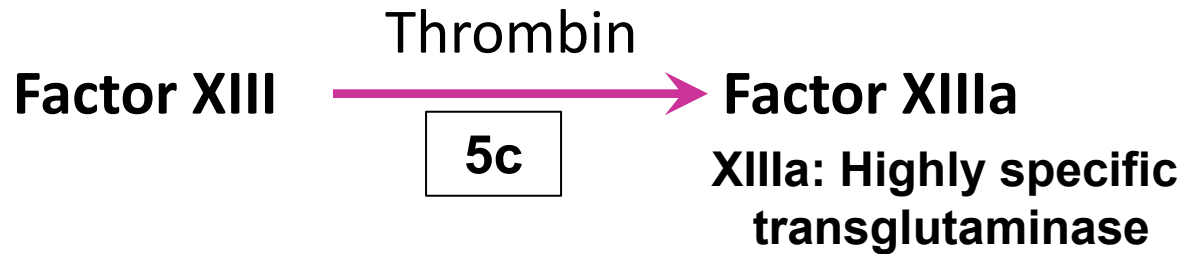


Medscape

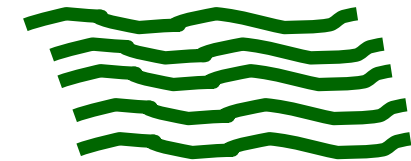
Source: Lab Med © 2017 American Society for Clinical Pathology

Covalent links

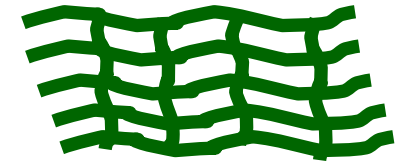
Formation of 'hard' clot mediated by Factor XIII activated by thrombin



Fibrin polymer (Soft clot)



XIIIa **6c**



Stable Fibrin (hard clot)

Covalent cross-linking of adjacent fibrin monomers to form stable cross-linked fibrin

Common pathway: Summary

1c - Thrombin activates factor V

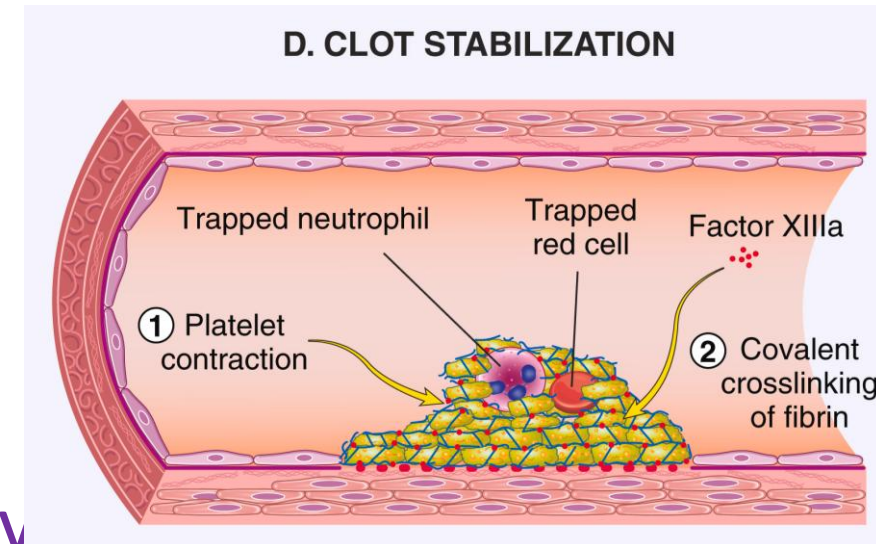
2c - Xa with Va, phospholipids and Ca^{2+} forms **prothrombinase complex**

3c - Prothrombinase complex converts prothrombin to **THROMBIN**

4c - Thrombin converts fibrinogen to FIBRIN

5c - Thrombin activates factor XIII

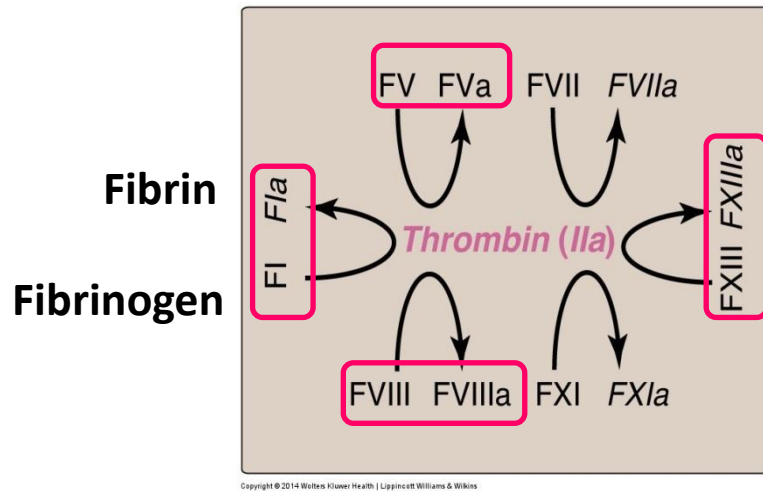
6c - Fibrin monomers covalently cross-linked by **factor XIIIa** to form cross-linked fibrin (**Hard clot**)





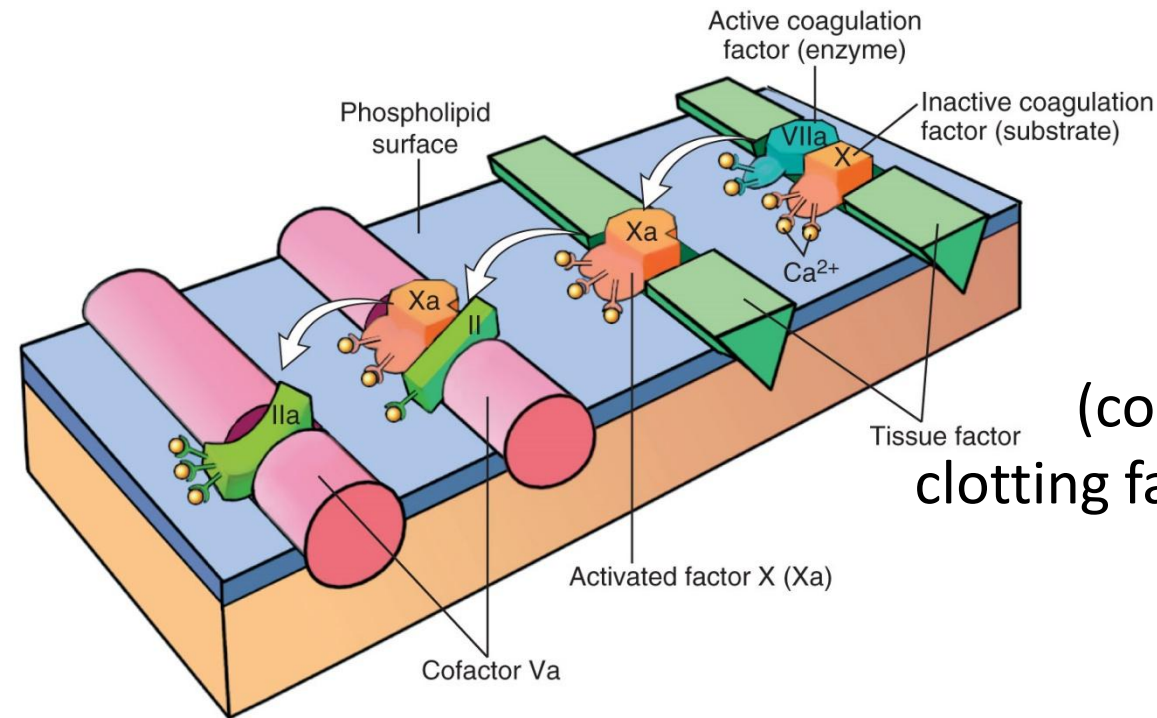
Role of Thrombin (Factor IIa)

- **Critical step** in coagulation cascade
- Activates Factor VIII and Factor V and further **amplifies** the coagulation cascade
- Converts **fibrinogen** (Factor I) to **fibrin**
- Helps in formation of 'stable thrombus' by activating **factor XIII**
- Thrombin activates **platelets**
- Thrombin activates **anticoagulant system**



Role of platelet phospholipid (Platelet plug) in clotting

Platelet activation exposes phospholipids on platelet surface



↓

Platelets facilitate secondary hemostasis (coagulation cascade); Provide binding sites for clotting factors. Ensures clotting occurs only at site of injury (platelet plug)

Factor IXa, VIIIa, PL, Ca²⁺: Activates **Factor X**
Factor Xa, Va, Ca²⁺, PL: **Prothrombinase complex** forms thrombin

Tests for coagulation disorders

Prothrombin time (PT) or International normalized ratio (INR)

- Tissue factor, phospholipids, and calcium added to blood and time for formation of fibrin clot is measured
- Tests **extrinsic** and common pathways
- Measures defects in
 - **Tissue factor**
 - **Factor VII**
 - Factor V
 - Factor X
 - Prothrombin
 - Fibrinogen

Activated Partial thromboplastin time (APTT/ aPTT/ PTT)

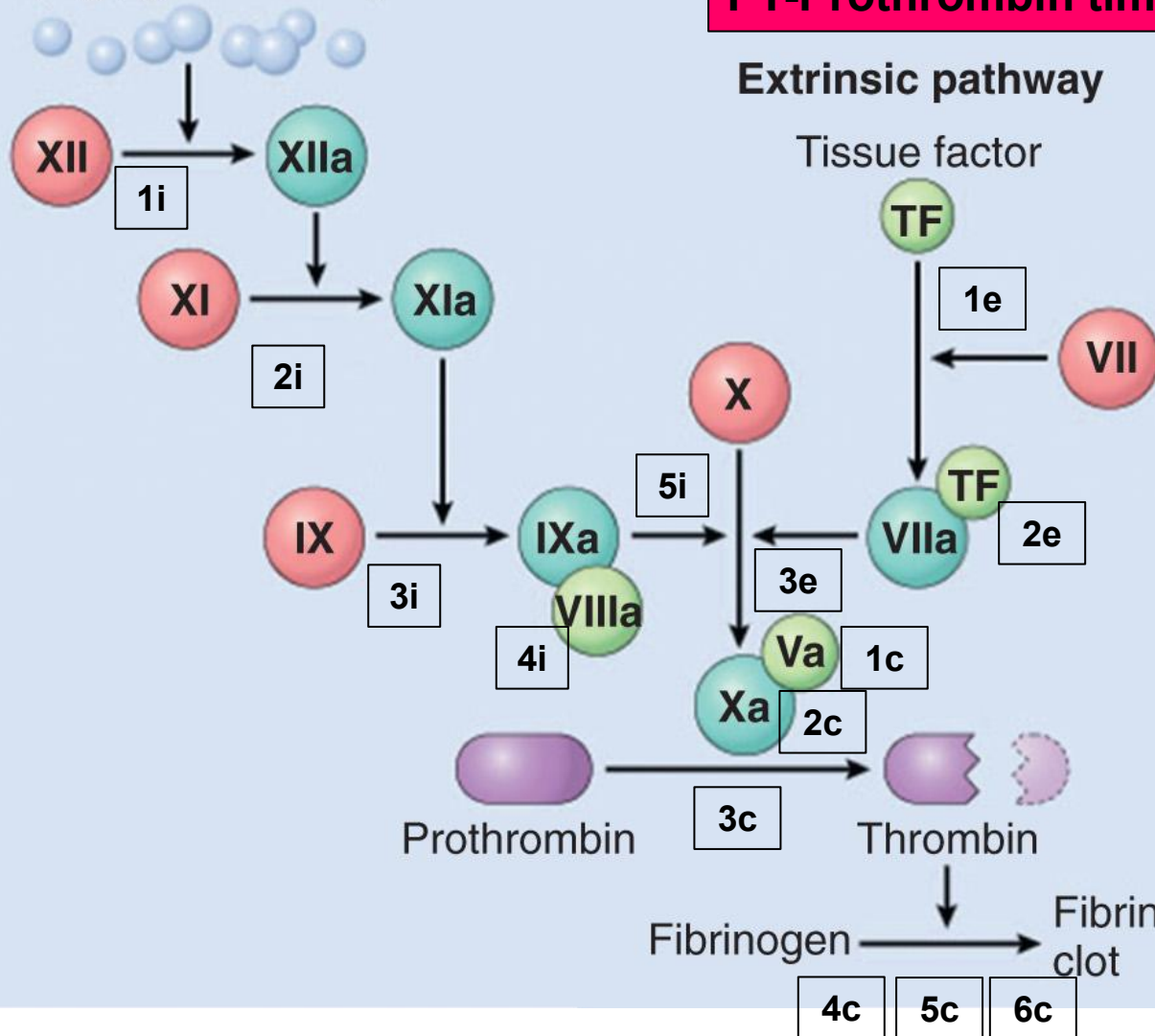
- Activate factor XII (glass beads) along with phospholipids and calcium, and time to form fibrin clot is measured
- Tests **intrinsic** and common pathways
- Measures defects in
 - **Factor XI**
 - **Factor XII**
 - **Factor VIII**
 - **Factor IX**
 - Factor V
 - Factor X
 - Prothrombin
 - Fibrinogen

APTT

CLOTTING IN THE LABORATORY

Intrinsic pathway

Negatively charged surface
(e.g., glass beads)



PT-Prothrombin time/ INR

Extrinsic pathway

Tissue factor

Factor I – Fibrinogen

Factor II- Prothrombin

Factor III-Tissue factor

Factor IV- Calcium

Factor V

Factor VII

Factor VIII

Factor IX

Factor X

Factor XI

Factor XII

Factor XIII

LEGEND:

Green – Common pathway factors

Blue – Extrinsic pathway factors

Black – Intrinsic pathway factors

Clotting factor song:

<http://www.youtube.com/watch?v=TG9yPHH3FmY>

Key Facts of Hemostasis

- Platelet plug formation requires: GPIb, GPIIb-IIIa, GPIa, vWF, platelets
 - Laboratory test for platelet plug formation: **Bleeding time; Platelet aggregation tests**
- Coagulation cascade:
 - Factor Xa generated by
 - Extrinsic pathway (1e-3e) – through VII and III (*activated on injury*)
 - Intrinsic pathway (1i-5i) – through XII, XI, IX and VIII
 - Factor Xa generates thrombin
 - Thrombin converts fibrinogen to fibrin (4c) and stabilizes fibrin (6c)
 - Laboratory tests for coagulation cascade:
 - Prothrombin time (INR): Extrinsic pathway
 - Activated Partial Thromboplastin time (aPTT): Intrinsic pathway

Proteins/ Enzymes required for platelet plug formation	Biological role	Disorder associated
GPIb	Binds activated platelet to subendothelial collagen via vWF	Bernard Soulier syndrome: Prolonged Bleeding time; Abnormal platelet aggregation
Von Willebrand factor (vWF)	- Binds activated platelet to subendothelial collagen - Stabilizes Factor VIII	von Willebrand disease: Prolonged bleeding time; Abnormal platelet aggregation and abnormal ristocetin cofactor assay and Prolonged APTT
GPIIb/IIIa	Platelet aggregation via fibrinogen	Glanzmann thrombasthenia: Prolonged bleeding time; Abnormal platelet aggregation
Platelets	Platelet plug formation	Thrombocytopenia; Prolonged bleeding time; Abnormal platelet aggregation

Proteins/ factors Secondary hemostasis	Biological role	Disorder associated
Tissue factor	Activates extrinsic pathway	
Factor VII	- Activated by Tissue factor - VII + Tissue factor activates factor X (extrinsic pathway) + calcium and phospholipids	Vitamin K deficiency Prolonged PT
Factors XII, XI	Activates intrinsic pathway	
Factor VIII	- Factors VIII and IX activate factor X (Intrinsic pathway) + calcium and phospholipids - Von Willebrand Factor stabilizes factor VIII - Inactivated by Proteins C and S	Hemophilia A: Prolonged APTT Also, in vWF deficiency (due to unstable factor VIII)
Factor IX	- Factors VIII and IX activate factor X (Intrinsic pathway) + calcium and phospholipids	Hemophilia B: Prolonged APTT
Factor X	- Convergence of Intrinsic and extrinsic coagulation cascade - Activates prothrombin to thrombin	
Factor V	- Factor X, V, calcium and phospholipids (Prothrombinase complex) - Inactivated by proteins C and S	
Thrombin (Factor II)	- Converts fibrinogen to fibrin - Activates Factor XIII (cross-links fibrin) - Activates factors V and VIII - Activates platelet plug - Activates thrombomodulin → Activates Protein C and S	
Fibrinogen	- Converted to fibrin by thrombin	
Factor XIII	- Converts fibrin (Soft clot) → Cross-linked fibrin (Stable clot)	
Calcium (Factor IV)	- Required for conversion of prothrombin to thrombin and factor X activation	